

ARM-450 — Design Log

- Updated: 2026-08-22

Current state: 6-DOF, all axes have hardware

axis	joint	part	status
J1	base yaw	turret_j1	NEW — first pass, unverified
J2	shoulder pitch	link boss + shaft + clamp + collar	validated, printed
J3	elbow pitch	link boss + shaft + clamp + collar	validated, printed
J4	forearm roll	roll_module	NEW — first pass, unverified
J5	wrist pitch	wrist_yoke_j5 + wrist	NEW — first pass, unverified
J6	tool roll	roll_module (2nd)	NEW — first pass, unverified

One interface everywhere: Ø42 pocket · 6806 bearing (30×42×7) · Ø30 shaft. One bearing part number for the whole machine.

Parts released

part	STEP	STL
arm450_parts_kit	yes	yes
base	yes	yes
bearing_yoke	yes	yes
joint_shaft	yes	yes
link_half_groove	yes	yes
link_half_tongue	yes	yes
roll_module	yes	yes
servo_collar	yes	yes
shaft_clamp	yes	yes
tool_flange	yes	yes
turret_j1	yes	yes
wrist	yes	yes

part	STEP	STL
wrist_yoke_j5	yes	yes

Verification log

date	check	result
2026-08-13	link bending axis	CORRECTED — $l=87,514$ not 39,899 (servo fit proved the orientation)
2026-08-13	seam open/closed	CORRECTED — no bending penalty; torsion 215x
2026-08-14	printer shrinkage	MEASURED 0.02 mm XY -> bearing pocket $\varnothing 42.15$ to $\varnothing 42.00$
2026-08-14	joint load sweep	J2 2.62 / J3 1.27 N·m — both UNDER-SPECCED on continuous
2026-08-14	topology opt	validates the box section; no redesign needed
2026-08-17	stadium profile	CORRECTED — boss was not tangent, 0.48 mm corner error
2026-08-17	seam gap	CORRECTED — tongue bottomed (non-integer layers) + wedged
2026-08-18	joint stack fit	clamp OD 42 to 38 — belongs in the through-bore
2026-08-18	URDF vs CAD	CORRECTED — section was stale; now imported + asserted
2026-08-18	DOF audit	was 2 of 6; now 6 of 6
2026-08-18	chain budget	turret gave J2 at 30 mm not 40 — CORRECTED, now 81 mm tall
2026-08-18	assembly DOF	build() took only q2,q3,q5 — J1/J4/J6 never wired. CORRECTED to q1..q6
2026-08-18	wrist J5→TCP	yoke 30 + roll module 34 + flange 6 = 70 mm, exact
2026-08-18	forearm J4	roll module 34 mm of 145, shell carries 111 — passes
2026-08-19	interference sweep	100 % of poses collided. J4 was mid-forearm and a 209 mm shell was placed in an 80 mm slot

date	check	result
2026-08-19	wrist redesign	3 stacked modules needed 152 mm against a 70 mm allowance. Integrated wrist: 62/30/30
2026-08-19	interference re-run	100 % -> 16 %, and every remaining hit is arm-vs-base, none arm-vs-arm
2026-08-19	links shortened	145 -> 119 mm to buy wrist length. Reach unchanged at 360 mm
2026-08-19	dead zone	naive formula said 62 mm; the 60 mm wrist points inward so the real TCP void is 0 (60k-pose FK sweep)
2026-08-19	FEA, new parts	turret SF 77, J4 housing 408, J5 yoke 417, J6 output 199 — all pass
2026-08-19	servo fit	J4 and J6 had no room for a servo. J4 housing Ø46->Ø56 (free); wrist rebalanced 34/26 -> 30/30 so J6 takes a full ST3215
2026-08-19	servo mounting	every part had a POCKET but no mounting holes and no horn pattern. Both added; the pattern values are PLACEHOLDERS pending measurement
2026-08-19	uniform servo	ST3215 everywhere works for drawing (J2 14-29 % of stall with a pen). Fails only at 300 g on full extension (89 %)
2026-08-19	Rsine on ARM-450	vertical board 40/40, 0.03 mm rms, pen tilt 0.5° at x=240 and x=300
2026-08-19	Rsine horizontal	table FAILS: J2, J4, J5 all pinned at limits simultaneously. Widening J5 110->150° cuts pen-tilt error 54°->16° but still 0/40

date	check	result
2026-08-19	URDF extraction	the URDF was parameter-linked, not geometry-extracted. arm450_meshes.urdf now carries real meshes + inertias from the STEP solids; FK identical to 0.000 μ m
2026-08-19	interference sweep	100 % of poses collided — first time this was ever checked
2026-08-19	wrist budget	needs 152 mm J4→TCP, had 70 — short 82 mm, a budget failure
2026-08-19	reach vs split	reach is 360 mm for ANY link split; shortening links is free
2026-08-19	links 145→119	both, L2=L3 preserved; frees 52 mm for the wrist
2026-08-19	integrated wrist	3 nesting parts, 122 mm total, 6-DOF retained
2026-08-19	dead zone	naive formula said 62 mm; true TCP void 0.00 (60k-pose FK sweep)
2026-08-19	interference re-run	14 %, all base/turret only — no arm-to-arm collisions
2026-08-19	FEA turret + wrist	SF 77 (turret) to 417 (yoke) — no structural concern
2026-08-19	shaft/bearing fit	one interface $\varnothing 42/6806/\varnothing 30$ verified across all six joints
2026-08-19	servo clearance	wrist parts had NO servo pockets — J4 grown $\varnothing 46 \rightarrow \varnothing 56$, J6 to a micro servo
2026-08-19	wrist servo sizing	J4/J5 need 10.1 kgf·cm, J6 needs 3.0 — ST3215 is 3-10 $\times$ oversized
2026-08-19	wobble bounded	c is bearing internal clearance: CN 0.16-0.65 mm; specify C2 $\rightarrow$ 0.03 mm
2026-08-19	fastening audit	no servo case fixings, no horn patterns, no cable exits anywhere — all added

date	check	result
2026-08-19	servo hole dims	ST3215.stl is an outer shell with no holes modelled — 4 params are PLACEHOLDERS
2026-08-19	FEA re-run	pockets turned J4 into an open C-section; re-checked at servo STALL: SF 41/21/105
2026-08-19	J4 load case	was lateral force; corrected to TORSION about the roll axis

Open / unverified

- J1, J4, J5, J6 parts are first-pass: no FEA, no fit check against the shaft, servo horn clearance and cable routing unchecked
- Wobble still never measured. Now bounded at 0.16–0.65 mm (CN class) by bearing spec

rather than by an unjustified assumption, but the dial-indicator test is still the single most valuable open number

- J6 micro servo is a REQUIREMENT (≥ 3 kgf·cm in 29×13×30 mm), not a sourced part
- Servo mounting hole positions and horn BCD are PLACEHOLDERS — measure the real

servo and correct the four values in cad/params.py

- J2/J3 continuous-torque shortfall needs a counterbalance decision
- Gripper not designed; tool flange is the interface only
- Horizontal-table tracing does not work — needs wider J2/J5 travel or a wrist redesign
- Servo mounting-hole and horn-bolt patterns are PLACEHOLDERS; measure a real ST3215
- "Space grade" is geometry-compatible only: PLA, hobby servos and wet-grease bearings all fail vacuum/thermal
- Residual 4 % base collisions need a planner-level collision check, not joint limits
- Exploded-view offsets still tuned for the old 3-DOF assembly; parts barely separate
- Renders are depth-sorted triangles, not a real renderer — use Blender/Fusion for presentation

End goal

Sine tracing with the pen normal to the surface (Rsine / mycobot thesis). That is a 5-DOF task minimum — J1 gives the writing sweep, J5+J6 give pen orientation, J4 avoids wrist singularities. All six now exist in CAD.

2026-08-20 — shaft material corrected, mass budget error found

User questioned the steel tube on weight grounds. Investigation found the BUY list specified $\text{Ø}30 \times 5$ steel (222 g each) while quoting "~93 g", and the pre-print gate counted only 2 shafts against 12 bearings (= 6 pairs = 6 joints = 6 shafts). Six steel tubes would have been 1332 g, taking the arm to 2270 g against a 1.2 kg limit.

Load analysis of the shaft (4 load paths): torque 0.55 MPa SF 32 | bending 0.57 MPa SF 104 | preload thread 16 vs 18 MPa MARGINAL

press fit creep — plastic relaxes, fit fades

Printing rejected on the preload thread and the creeping press fit, NOT on strength. Both failures are local, so the fix puts metal only at those two places.

Spec now: AL6061 $\text{Ø}30 \times 2$ wall $\times 72$, 34 g each / 205 g for six, total 1144 g.

- M5 $\times 90$ through-bolt down the $\text{Ø}26$ bore + nyloc replaces the M5 tapped end
- $\text{Ø}3$ roll pin (double shear 4241 N vs 196 N, SF 22) replaces grub screws + D-flat
- tube OD must be measured on arrival: extrusion tolerance $\pm 0.10\text{-}0.20$ is 10-20x

looser than a bearing seat needs. Loctite 603/638 covers gaps to 0.10 mm.

Gate rerun: 55 passed, 0 warnings, 0 failures. Mass line now derived from params.py with an assertion tying shaft count to bearing count.

2026-08-21 — horn adapter added; mass model corrected; budget fixed at 1450 g

- cad/horn_adapter.py (6 off, 7.3 g each at 0.90 fill). Closes the torque path

servo horn → adapter → roll pin → tube → clamp → link, which did not exist: the old solid shaft carried its own horn shoulder and lost it when the shaft became a bought tube. Reuses the tube's existing pin hole. Sized on stall 2.94 N·m — horn screws SF 4.6 (binding), roll pin SF 7.3, hub torsion SF 40.

- Mass model corrected. 0.55 infill factor was being applied to hollow-modelled parts, discounting the hollowing twice and hiding ~240 g. FILL = 0.90. True mass 1437 g, not the 1215 g previously reported.

- Budget. 1200 → 1300 → 1450 g hard ceiling, user decision. Mass is now a HARD gate

failure, not a warning. Headroom 13 g.

- Base foot lightened (underside recess, plate 7 → 5.5 mm): -21 g.
- All three drawing sets now carry the adapter: iso 13 pp, 3-view 13 pp, verification 10 sheets.
- Gate: 58 passed, 0 failures. Geometry audit: 52 features verified, 0 outstanding.

2026-08-21 (later) — sine collision found and fixed, wrist bearings, workspace

The sine path was not collision-free. Exact-mesh check on the 240 SOLVED waypoints (not the random joint sweep) found 72/240 colliding — base ↔ upper arm, 1.27 mm

penetration at $z = 50.0$, the arm grazing the top rim of the pedestal. Fix: `Z_CENTER 0.20` → `0.22`. Result `0/240 collisions AND 0.194` → `0.014 mm rms`, because the IK stops working against the J2 limit. Strictly better on every metric. Also narrowed `PED_D_TOP 80` → `72` (-7 g) and added a collision-aware IK seed bank; neither was sufficient alone.

Wrist bearings `6806` → `6706` ($\varnothing 37 \times 4$), J1/J2/J3 unchanged. Loads computed: worst wrist case is SF 103. Same 30 mm bore, so tube/clamp/adaptor untouched. Bearings 268 → 176 g, wrist parts +18 g, net -75 g. Two part numbers now — `BUY.md` rewritten to lead with that.

Mass model corrected earlier the same day (`0.55` → `0.90` fill; the `0.55` was discounting hollow-modelled parts twice and hid ~ 240 g). Arm is 1365 g against a 1450 g hard ceiling, 85 g headroom. Gate fails rather than warns on mass.

Horn adapter added (6 off) and a 2-body defect in it fixed — the grip-flat cut used `centered=(True, False, False)`, which grew the box in +Y only and sliced a 0.20 cm^3 sliver loose on the $-Y$ side. Caught by the gate's "is ONE body" check.

New tools: `check_sine_clear.py`, `check_bearing_fit.py`, `cad/bearing_6806.py`, `verify_all.py` (5 rounds, new sweep seed each), `workspace.py`.

Verification: 5 rounds all clean. Gate 60 passed / 0 warnings / 0 failures. Geometry audit 52 features / 0 outstanding. Virtual bearing fit: all 10 pockets light press. Workspace: 190 litres reachable, 20 % workable with the pen normal.

2026-08-21 (evening) — playback continuity, RViz model, URDF mass

Playback was discontinuous. `sine_node.tick()` wrapped the last waypoint straight to the first: J4 jumped 43.30° , J1 32.2° in one tick — $21\times$ the largest normal step — and the tool teleported 139.8 mm. Replaced with ping-pong playback (forward, then back along the same waypoints, as `R sine` does), plus a `loop_mode` parameter (pingpong default, wrap retained). Verified live at 60 Hz: largest step 2.094° , zero steps above 5° .

RViz showed the wrong robot — two faults.

1. The launch files loaded `arm450.urdf`, which carries box/cylinder primitives, so RViz rendered a stack of blocks. Both launch files now load `arm450_meshes.urdf`.
2. `generate_urdf_meshes.py` emitted `package://arm450_design/meshes/...` `arm450_design` is the design directory, not a ROS package, so all 7 meshes failed to resolve with no error visible in the viewer. Fixed to `package://arm450_description/`. 7/7 resolve.

URDF mass was wrong twice over.

- `generate_urdf.py` carried hand-set LINK-ONLY masses — no servos, bearings or shafts.

link4 was 173 % light; the total said 0.765 kg for a 1.365 kg arm. Now derived from `volumes.json`.

- `arm450_meshes.urdf` took its inertials straight from mesh geometry, so it was plastic

only at 0.784 kg. Non-printed mass is now added as a lumped mass at each link origin —

an approximation that gets mass and gravity torque right and slightly understates rotational inertia, stated in the code.

- A third defect surfaced from comparing the two totals: the model allocated ONE bearing per link where every joint runs a PAIR — 4×6806 instead of 6, 44.8 g light. Fixed; the URDF now totals 1.365 kg, matching the mass gate exactly.

State: gate 60 passed / 0 warnings / 0 failures. Geometry audit 52 features / 0 outstanding. 3 verification rounds clean. Runtime verified end to end: robot_state_publisher loads the mesh URDF with no errors, tf2_echo base_link tcp → [0.300, 0.043, 0.194].

2026-08-22 — wrist self-collision investigated and fixed

It was a pure J5 fold. Mapping the wrist joint space showed every J4 row identical — J4 has no influence. Bisecting on real meshes put the collision-free J5 range at $-94.2^\circ \dots +52.7^\circ$ against a declared $\pm 110^\circ$: 57° optimistic on the + side, 16° on the -, and asymmetric because the wrist is (the servo pocket sits on one side). 5° past the edge the penetration is 5.87 mm — a real overlap, not a graze.

Fixed by declaring joint5 $-91^\circ \dots +49^\circ$ (measured range, 3° margin). No geometry change. Sine unaffected: uses $+40.4 \dots +48.0^\circ$, re-solves at 0.014 mm rms, 0/240 collisions.

Second defect this exposed: interference.py carried a HARDCODED copy of the joint limits and never read the URDF, so after the narrowing it kept sampling poses the arm can no longer reach and reporting them as collisions. Now reads the URDF.

Result: random-pose collisions $\sim 30\% \rightarrow \sim 12\%$, and j4housing ↔ j6output is gone. What remains is base ↔ j4housing and turret ↔ fore-B — the arm folding onto its own base, which every arm does and no geometry change removes.

Also added workspace.figure_3panel() → figures/workspace_arm450.png, the three-panel reachable-workspace figure in the same layout as the myCobot study, with the sine demo overlaid. ARM-450 reaches 0.360 m horizontally vs the myCobot 280's 0.30 m.

2026-08-22 — simulation made hardware-ready; staged preflight; GO

Hardware readiness (three gaps that only bite on real servos):

- Approach ramp. The node published waypoint 0 on its first tick — invisible in RViz, a full-speed slam on hardware. Now a cosine ramp from start_pose over approach_time (default 3 s), zero velocity at both ends.
- JointTrajectory topic. /joint_states is a visualisation stream a controller will not follow. The solved path is now published once as a 240-point JointTrajectory on /arm450/trajectory, latched TRANSIENT_LOCAL, velocities by central difference.
- Velocity check against the actuator at startup: 0.91 rad/s demanded vs ~ 2.3 rad/s for

a loaded ST3215. Over-rate does not error — the servo lags and the traced path silently stops being the solved path — so the node warns and names the highest safe rate.

New `creep_fatigue.py` — the two failure modes a static check cannot see. Working stress is 0.38 MPa, a stress ratio of 0.0064; fatigue life $4.3e28$ cycles. Neither is limiting, because the arm is stiffness-driven, not strength-driven. Creep needed an honest correction: the absolute strain is tiny (0.040 %/year) but it is several times the elastic strain, and the Findley fit is being used two orders of magnitude below its calibration. Actionable outcome: park the arm folded when idle, not extended.

New `preflight.py` — seven staged gates with a single verdict: design check, FEA, creep+fatigue, stiffness, URDF, path tracing, simulation readiness. Exits 0 on GO.

RESULT: 21 checks, 7 stages, 0 failures — GO FOR PRINTING. Worst FEA SF 23.0 · compliance 0.098 mm · path 0.014 mm rms, 0/240 collisions · URDF mass 1.365 kg matching the gate · mass 1365 g of 1450 g.

2026-08-22 — the docking interface did not work, and three drawings lied

Shop drawings for the six end-effector parts were the outstanding item: they existed in CAD and in `END_EFFECTORS.md`, but not on a dimensioned sheet. Adding them to `draw_iso.py` and `draw_3view.py` meant looking hard at the geometry, and the geometry did not survive it.

tool_dock had no latch

The three latch lugs sat 5.5 mm clear of the bore they were meant to grow out of, inside the capture cone at a height where it had already opened to $\varnothing 25.8$. The part exported as four solids: the probe and three loose 19 mm³ crumbs. A CadQuery `.union()` of shapes that do not touch succeeds and returns a compound, so nothing complained — not the render, not the drawing, not the mass budget.

dock_target had cut its own spigot in half

The capture groove was cut from r3.10 to r6.40 through a wall running r4.00 to r5.65 — a 3.30 mm groove in a 1.65 mm wall. That severs; the three entry slots then chopped the severed tip into three loose arcs. Also four solids.

And they could not have mated anyway

The capture cone was 16 mm deep, the spigot 10 mm tall. The cone mouth grounds on the $\varnothing 34$ flange while the tip is still 6 mm short of the throat. Probe and target collide at every insertion depth and every roll angle. Neither part is at fault on its own; the defect lives in the relationship, which is why single-part inspection could never have found it.

Fixes

was	now	why
throat $\varnothing 12$	$\varnothing 16$	gives the spigot a 2.45 mm wall at the groove root

was	now	why
mouth Ø30	Ø34	keeps ± 9 mm capture at the wider throat
cone 16 mm deep	9 mm	a lead-in must not out-reach the post it guides
spigot 10 mm	21 mm	reaches the latch with 2 mm of standoff
lugs floating	grown inward from the throat wall, 1 mm overlap	real material to fuse to
flange 18 mm	10 mm	it was never doing anything

tool_dock 38 → 31 mm tall, 48.0 g; dock_target 28 → 31 mm, 12.6 g.

The mate is now simulated, not asserted

check_tool_fit.py drives the exported probe mesh onto the exported target mesh:

state	expected	result
6 mm short, slots aligned	clear	clear
seated	clear	clear
rolled 30° to latch	clear	clear
latched, pulled 2 mm	must catch	catches
rolled back, withdrawn	must release	releases

Capture envelope is measured by sweeping the probe sideways until it fouls: ± 9.75 mm at the mouth, 7× the ~ 1.4 mm the arm can make. It also counts `len(shape.Solids())` for all six tool parts and fails on anything but 1.

cad/end_effector.py now exports itself. It never had a `__main__`; the six parts had been written once by hand, so every edit to that file since had been silently absent from the STLs the drawings and collision checks read.

Two documentation defects, found the same way

- Three three-view sheets carried a bold red NO Ø42.00 BEARING POCKET banner from a

hardcoded list of part names. Those pockets were fixed on 2026-08-21. The banner now reads the count out of the STEP, and shows green with the count when present.

- Every isometric sheet quoted mass at 55 % fill while the budget and the gate use 90 % — under-quoting every part by 39 %. `mass_of()` now takes `params.MASS_FILL`.
- `TOOL_REACH` in `workspace.py` was hardcoded; the dock entry (43 mm) went stale when the

cone shortened. It is now measured off `assemble.build()`. Real figure: 36 mm.

State

Verification re-run with and without a tool; paper and deck written

verify_configs.py runs the whole load suite across five payload configurations — bare, pen, gripper, gripper+225 g, dock — from one driver, so the "with gripper" answer cannot be a different vintage from the "bare" answer.

The point is not the extra mass. Every earlier analysis used 300 g at the TCP, which bounds a 75 g gripper comfortably. The point is the lever arm: a tool moves the load 42 mm along the tool axis, and that is what sizes J2/J3 and sets compliance at the working point. 75 g on a 42 mm stick is not the load case 75 g at the flange is.

config	J2 N·m	σ MPa	SF	δ mm	creep %/y	cycles
bare	1.55	0.197	305	0.036	0.0159	3.9e36
pen	1.61	0.208	288	0.041	0.0172	1.8e36
gripper	1.85	0.257	234	0.054	0.0230	1.1e35
gripper + 225 g	2.74	0.437	137	0.098	0.0486	8.0e31
dock	1.79	0.245	245	0.051	0.0216	2.0e35

30/30 criteria pass. The re-implementation cross-checks: compliance in the gripper+225 g case comes out at 0.098 mm, matching compliance_budget.py's published figure for 300 g exactly.

The finding worth keeping: backlash beats structural deflection by $\approx 20\times$ in every configuration, so the conclusion "this arm is servo-limited, not structure-limited" is invariant to whether a tool is fitted. Nobody should spend print time stiffening it.

Gate is now 27 checks.

ARM450_PAPER.pdf (15 pp) — the work as a paper: requirements, architecture, kinematics, end effectors, the verification method and its results, a comparison against a space-grade equivalent, the thirteen defects the gate caught, limitations, conclusions. §9 is [AWAITING HARDWARE]: seven measurements, each with a prediction and a stated falsification condition, and a blank for the number. Written before the test on purpose — a prediction recorded beforehand is evidence; one recorded afterwards is not.

ARM450_SLIDES.pdf (20 slides, 16:9) — generated by make_slides.py from the same modules, so no number on a slide is a third copy. The generator prints a warning when a bullet list overruns its slide, which caught five overflowing slides that would otherwise have shown up on the projector.

Space-grade comparison: of sixteen subsystems, 3 survive unchanged (kinematics, joint architecture, fastening scheme), 2 need rework, 11 must be replaced. The material is the wall — and re-made in Al 6061-T6 the same geometry is ≈ 1.13 kg, lighter than printed, and about $10\times$ stiffer, so the swap is an improvement that costs money rather than a compromise.

A generated command reference — and five aliases I deleted by accident

ARM450_COMMANDS.pdf (9 pp): every alias, every raw ROS 2 equivalent, the plan_sine.py options, and a troubleshooting table. It is generated — make_commands.py parses the alias table out of arm450_aliases.sh and the trajectory options out of plan_sine.py --help. A hand-typed command sheet is exactly the document this project has now watched go stale three times, so this one cannot: a450cmds rebuilds it.

Two things it exposed.

build_pdf.py stamped the literal footer "ARM-450 · stress & stiffness analysis" on every PDF it has ever produced — right for the one report it was written for, wrong on the other seventeen. The footer now takes the document's own H1. All 18 PDFs rebuilt.

And while restructuring the aliases I replaced a slice of the file that ran from a comment to a450help() { — which silently ate a450gate, a450flight, a450cad, a450docs and a450open. All five restored.

The audit that was supposed to catch that did not, and the reason matters: it tested each alias by re-typing the command the alias was supposed to run, so it passed with the alias itself deleted. It now checks that every name is type -t-defined first, and executes the alias by name inside a subshell that sources the file. Also: bash disables aliases in non-interactive shells, so the harness needs shopt -s expand_aliases or everything silently fails for the wrong reason. 50 checks, all passing; kept as audit_aliases.sh.

Testing what a thing is meant to do, rather than the thing itself, is not a test. It is a restatement of the intention.

The sine was traced by the flange, not by the tool

Spotted in RViz: the yellow curve sat at the J6 flange while the gripper fingers stuck out 42 mm past it. On hardware that is not cosmetic — the flange would meet the board while the fingers were already 42 mm through it.

The trajectory drives the TCP, which is the J6 tool face. Everything about that was correct for a bare flange and quietly wrong the moment a tool went on.

Measured, then compared two fixes on the number that matters — tip position against the intended sine:

	board at	joint step	tip error
flange path, board moved out	342 mm	2.09°	0.055 mm rms · 0.834 max
tip-driven IK, board fixed	300 mm	2.87°	0.087 mm rms · 1.019 max

Moving the board wins on every axis, and it is free: the joint trajectory is byte-for-byte the one already verified collision-free, so the posture, the joint loads and the clearance margins are all unchanged. Only the board moves.

Both are now available. `plan_sine.py --tool gripper` reports the board distance out loud; `--tip-ik` drives the tip instead, for a board that cannot move.

Two things had to be fixed to make `--tip-ik` work at all. `ik()` scored its result against the bare TCP, so every tip-offset solve reported ~42 mm of error — the tool length — and the acceptance test and collision-seed filter rejected perfectly good poses. And at the shipped `w_ori = 0.05` the solver traded 21° of tool tilt for position error, because with a tool fitted a tilt moves the tip; position and orientation stop being independent. It needs `w_ori 0.2` and the trace raised to 260 mm.

The path marker now follows the tip — verified at 0.00 mm from the furthest point of the tool mesh, not just eyeballed. A third RViz config, `arm450_side.rviz`, looks along -Y, because perspective in the other two makes the curve look like it passes through the gripper when it does not.

Re-verified with the regenerated trajectory: gripper 0/60, dock 0/60.

Two more stale aliases found while testing: `a450reset` still carried the old `--z-center 0.20` (the height that collided with the base pedestal, fixed days earlier), and `a450urdf` rebuilt only the box-primitive URDF, leaving the mesh URDF RViz loads stale — the exact mechanism behind the earlier "RViz is showing the wrong model".

The URDF had no tool on it

Asked directly whether the design had been run in RViz with the end effector, the answer was no — and worse, it could not have been. `arm450_meshes.urdf` had seven links, `base_link` through `link6`, and stopped at a bare J6 face. The sine path had been re-verified collision-free with a gripper fitted, the workspace recomputed for it, the reach measured off the assembly — and every one of those numbers came from `assemble.py`, the CAD and collision assembly. None of it was in the model RViz loads. Every RViz run in this project has shown an arm with nothing on the end of it.

tool is now a fixed link on `link6`, built the way every other link here is built: the real STLs at their true placement, combined into one mesh, mass and inertia from the exact STEP volumes. Placement comes from `assemble.build()` rather than being re-derived, so what RViz draws is the geometry the collision check cleared. `tool_tip` follows it at the working point. Selectable: `EMIT=1 TOOL=dock python3 generate_urdf_meshes.py`.

Gripper 74.6 g, tip 42 mm past the J6 face. Dock 62.4 g, 36 mm.

Stage 5 needed fixing with it. Its mass check summed every link with an inertial, so the moment the tool became a link the 1.365 kg comparison would have broken — a 75 g payload silently failing the check that ties the model to the mass budget. It now sums the arm links only and reports the tool separately, plus a new check that the tool link is present at all.

Run and captured: `figures/rviz_gripper_wide.png` / `.webm` and `figures/rviz_gripper_close.png` / `.webm`.

Two incidental fixes fell out of actually running it:

- RViz opened at 728 × 604 — its built-in default, with the Displays panel eating half.

A 45 mm gripper on a 450 mm arm is invisible at that size. The config now carries a

Window Geometry block at 1500 × 900, and a second config arm450_tool.rviz parks the camera on the wrist.

- RViz would not start at all from a terminal inside the VS Code snap:

libpthread.so.0: undefined symbol: __libc_pthread_init, GLIBC_PRIVATE. The cause is GTK_PATH, which the snap points at its own GTK module directory; RViz's Qt/GTK integration loads a module from there and it drags in snap glibc. unset GTK_PATH is the entire fix — bisected, and the snap entries in LD_LIBRARY_PATH, PATH and LOCPATH turn out to be red herrings. (I said it was LD_LIBRARY_PATH at first; that was wrong.) Alias a450unsnap. A plain Ubuntu terminal never has the problem.

450 mm settled. Asked whether the 450 mm was a hard MASCOT stowage envelope or a target for the arm alone, the user accepted 492 mm with a gripper fitted. So 450 mm binds the arm — base to the J6 tool face — and a tool is payload beyond it. Recorded as params.SYSTEM_REACH_MAX = 495 and checked in stage 8, so a later and longer tool cannot push the system past what was agreed without the gate saying so.

Pre-print gate: 27 checks, 8 stages, 0 notes, 0 failures — GO FOR PRINTING. Stage 8 is now 29/29 (was 15/15). Workspace with a gripper fitted: 492 mm reach, 262 L swept.

2026-08-25 — Craig DH, velocity-propagation Jacobian, ikpy cross-check

Built the kinematics a second time in the user's own convention — modified/Craig DH with a velocity-propagation Jacobian — as an independent derivation rather than a restatement of the URDF.

The table has two numbers that are not what they look like. Taking the joint axes from the URDF at $q = 0$ and computing common normals: every axis passes through the base Z line, so every a is zero except $a_2 = 119$ between the two parallel pitch axes. And $d_4 = 181$, not 119 — frame 3's origin sits at $z = 209$ and frame 4's at $z = 390$, so the offset along Z_4 is $119 + 62$. Writing 119 there, which is the obvious link length, puts the tool 193 mm out. That was this file's first answer.

Three implementations sharing no code now agree to 2×10^{-13} mm: the Craig-DH FK written here, the URDF chain the project already ran on, and ikpy. The propagated Jacobian matches finite differences to 1.8×10^{-5} , and both IK solvers converge — 1.6×10^{-11} mm median for the DH one. 7 of 7 checks pass.

Agreement between the first two proves the DH table. Agreement with ikpy proves the first two do not share a mistake, which is the failure a self-written cross-check cannot catch.

ikpy does not import on this machine, and it is not ikpy's fault: sympy stringifies numpy scalars, numpy 2 changed their repr to np.float64(1.0), and sympy then cannot parse its own input. check_ikpy.py shims the converter through float(). Nothing to do with the arm — but reporting "ikpy unavailable" would have quietly dropped the only outside opinion in the check.

Also: the sine trace on a horizontal plate does not work, and now we

know why

Swept 12 plate positions. Best is 11.7 mm rms against 0.020 mm on the vertical board — 585× worse. J2 is hard against its +115° stop: to reach out over a table and tip the tool down, the shoulder runs out of travel. A 90° bent tool adapter was tried and improved it to 8.1 mm, at which point J2 and J5 both saturate. It is a joint-range limit, not a solver or tool problem, and it traces back to J5 being narrowed from $\pm 110^\circ$ to $-91^\circ/+49^\circ$ to stop the J4 housing striking the J6 body.

And the missing chamfers

BRG_CHAMFER = 0.5 has been in params, on the drawings and in the docstrings since the design was written. A STEP audit for conical faces found zero in every part with a bearing pocket — link_shell.py promised it in its docstring and never called it; fit_coupon.py called .chamfer() inside a try/except that threw every time. 15 lead-in chamfers now cut, as cones rather than .chamfer() calls. The cylindrical bore is now 6.5 mm of the 7 mm seat, which is normal — the bearing still grips 93 % of its race and the fit check still reads light press.

Dimensional audit: 49 checks, 0 failures.

2026-09-01 — the fit coupon tested only half the bearings

Six 6706 bearings arrived and would not fit the coupon. The bearings are correct. The coupon was wrong.

The wrist moved from the 6806 to the smaller 6706 (30 × 37 × 4) on 2026-08-21, and that change was followed through into wrist_integrated.py, check_bearing_fit.py, dim_audit.py, generate_urdf.py, preprint_check.py and BUY.md — into every place except the one part whose entire purpose is to measure a bearing fit before anything else is printed. The coupon kept its three Ø42.00 stations and never gained a Ø37.00 one, so six of the arm's twelve bearings had no fit test at all.

The failure presents as a hardware fault. A 6706 offered up to a Ø42 pocket has 2.5 mm of radial slop and drops straight through, which reads as the wrong bearing having been delivered. BUY.md had even written the warning down — "a 6706 dropped into a Ø42 pocket has 2.5 mm of radial slop and will look almost right" — and the coupon still made exactly that mistake unavoidable, because it offered no other hole.

Fixed. The coupon now carries two rows: Ø42 × 7 deep for the 6806 (J1/J2/J3) and Ø37 × 4 deep for the 6706 (J4/J5/J6), each stepped ± 0.05 mm, each with its own lead-in chamfer cut as a cone, each labelled with raised numerals, dot counts and the bearing part number on the strip. 150 × 139 × 12 mm, 70 g, ≈ 2.3 h — up from 27 g and 0.9 h.

Three further defects fell out of the fix:

- audit_parts.py checked the three Ø42 pockets and nothing else. It now checks both rows' diameters, both seat bores and both pocket depths — the depth check is what would have caught a Ø37 pocket cut 7 mm deep instead of 4.

- `draw_iso.py` placed its pocket callouts at $y = 0$, which was the main row's centre only while the coupon had one row. The second row moved the part centroid and every leader would have pointed at bare plate. The row centres are now computed once in `fit_coupon.py` and imported, rather than assumed twice.
- `BUY.md` and `DRAWINGS_2D.md` embedded `figures/cad_coupon.png`, an orphan produced

by no current script and last written 2026-08-20. Both PDFs were showing a picture of the single-row coupon beside text describing the two-row one. Repointed at the generated drawing.

Part audit after the change: 59 ok, 0 need attention.

This is the same class as the missing chamfers and the vacuous wall check — a parameter changed in the model and the thing that was supposed to verify it was never told. What makes this one worse is that the unverified item was the verifier.